2.2.5.9 Azoic Dyes or Napthol Dyes

Unlike other colouring matters, the azoic dyes are not prepared as dye stuffs but have to be produced directly in the fibre by the combination of their constituents. The first component, napthol in presence of sodium hydroxide give colourless or faintly yellow or brown colour when applied on cotton at room temperature. The naptholated fabric when immersed in the second bath containing the second component base along with the acid and sodium nitrite at ice cold temperature produces bright colour which is insoluble in water. The two components form azo group during coupling. So the dye is known as **azoic dye**. Since it is formed inside the fibre, it is called **ingrain dye.** Azo dyes are also called as **ice colours** since their application involves use of ice to lower the temperature. They are widely used on cellulosic fibres and have limited application on acrylic, nylon and polyester. Azo dyes exhibit bright colours. They have good colour fastness to washing and light.

2.2.5.10 Pigment Dyes

Pigments are not true dyes because they have no affinity for the fibre. They are applied and held to the fabric with the help of adhesives and resins. Pigments are commonly used in dope dyeing and printing. Use of pigments saves both time and cost. Vat dye is called as dye when applied on fabric in solubilized form and is termed as pigment when applied on fabric in insoluble form using a binder. Pigment dyes exhibit excellent light fastness.

2.2.5.11 Optical Brightners

These dyes are also called as colourless dyes or fluorescent whiteners. These dyes absorb light at UV-region and emit blue light in visible region. It may be applied during bleaching or with the final finish. These dyes have affinity for cotton but are also used for wool, nylon, acetate and acrylics.

2.3 Dyeing Methods

The art of dyeing dates back to pre-historic times. It is practised from ancient time and it is as old as human civilization. Dyeing is the process of imparting colour to the fibre/yarn or fabric by the application of dyes or pigments. Colour is produced between the dye molecule and the fibre or fabric. The bond between dye molecule and fibre may be strong or weak depending on the dye used. Dyeing enhances the aesthetic look of the fabric. It is one of the surface ornamentation methods.

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2.3.1 Stages of Dyeing

Dyeing can be done at any stage of textile production process such as at fibre stage, yarn stage, fabric or finished product.

Table 2.4 Stages of Textile Dyeing

Stage	Method of dyeing
Fibre	Stock dyeing
Wool sliver	Top dyeing
Manmade	Dope dyeing
Yarn	Skein dyeing, Package dyeing, Warp-beam dyeing, Space dyeing
Fabric	Beam dyeing, Beck/Winch dyeing, Jig dyeing, Jet dyeing, Pad dyeing
Finished product	Product dyeing
Blended fabrics	Union dyeing, Cross dyeing

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